Cells and Gauge

File: gauge.kicad_sch

3V3 Rail

File: power.kicad_sch

MCU

File: mcu.kicad_sch

USB Port 1

IRQD
IRQ_N
EN_5V
EN_5V1
ADCIN1
P1_LDO_3V3
ADCIN2
LDO_3V3D
P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQD
IRQ_N
EN_5V
EN_5V2
ADCIN1
GND
ADCIN2
LDO_3V3D
GND

File: usb_port.kicad_sch

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: ADCIN1 tied to its LDO_3V3, ADCIN2 to GND  -> I2C 0x20
Port 2: ADCIN1 and ADCIN2 to GND                   -> I2C 0x21
Same SDA/SCL bus (CC2340 I2C0, 100 kHz): gauge BQ40Z50-R2 0x0B

MONOLITH PRODUCTION BOARD - SHEET INDEX
2.  Cells and Gauge:  welded 4S pack pads, BQ40Z50-R2 gauge, protection FETs, BQ294700 + F102 second-level protection, SMBus pogo pads
3.  3V3 Rail:  TPS62177 always-on 3V3
4.  MCU:  CC2340R5MODA, Tag-Connect TC2030-NL SWD pads
5.  USB Port 1  /  6.  USB Port 2  (one sheet file):  TPS25751D + TPD4S201 + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (gauge VSS only)
Spec and open items: Production/README.md.  Derived from the dev board rev A8 (DevBoard/, history in DevBoard/README.md).

REVISION LOG
P0  2026-10-17  Dev board rev A8 with the dev-only parts removed (Production/tools/make_production.py, tidy_production.py):
                sleep-current section, bench supply, cell holders, headers, buttons, status LED, test points, DNP options
                and OR links. Added pack wire pads, gauge pogo pads, TC2030-NL SWD pads, factory test pads.
                Author: Claude Code.  Reviewer: none yet.
Dev board history (A0-A8): DevBoard/README.md and the DevBoard root sheet.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Sheet index, PD address straps
**Monolith 100 W battery bank — production**

Sheet: /
File: Monolith.kicad_sch

***Title: Monolith production board — top level***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-17 | **Rev: P0** |
| KiCad E.D.A. 10.0.7 | | Id: 1/6 |

# PROTECTION FETs (back-to-back N-FETs, common drain)

## PACK WIRES  (welded 4S EVE INR21700-58E)

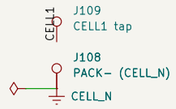
*J108 PACK- and J112 PACK+: 16 AWG leads in 1.7 mm plated holes. J109-J111: cell tap wires.*
*Connect with USB unplugged, in order: PACK-, CELL1, CELL2, CELL3, PACK+ (note 15).*

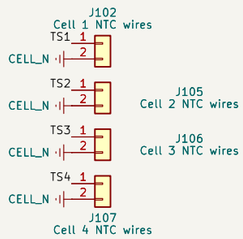


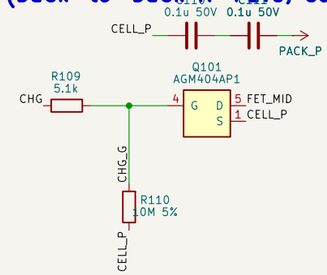


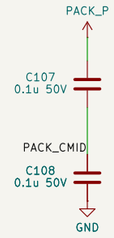




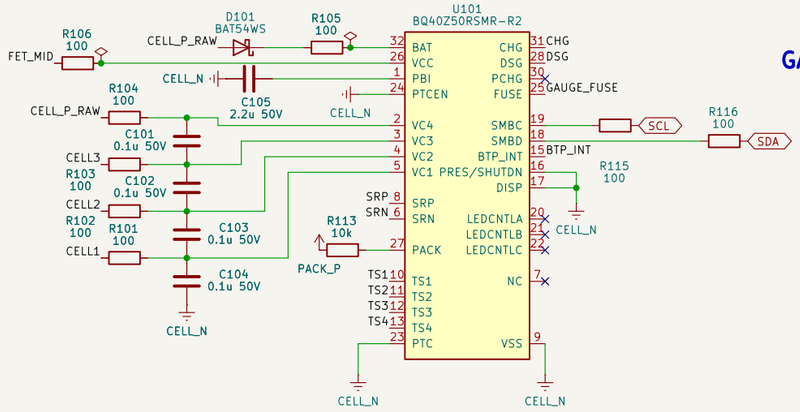



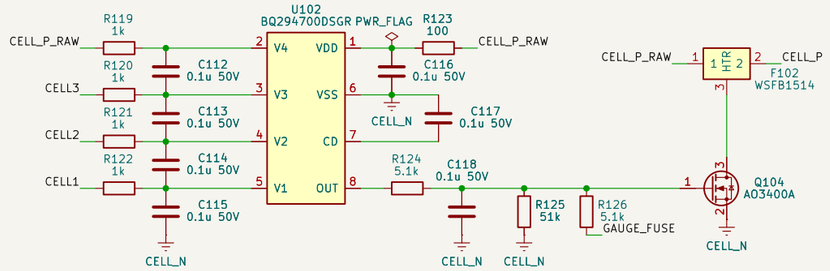



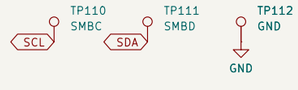

## GAS GAUGE  BQ40Z50-R2  (TI fig. 9-1)

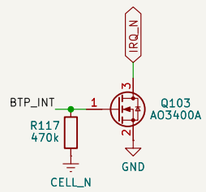



## SECOND-LEVEL PROTECTION  BQ294700 + SCP  (TI fig. 9-11)

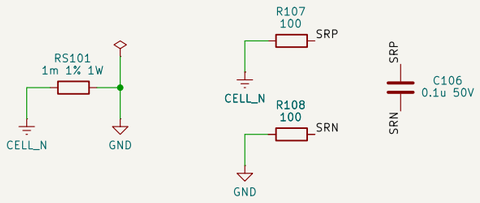



## GAUGE SMBus (0x0B) + POGO PADS




## LOW-SIDE CURRENT SENSE




## FACTORY TEST PADS

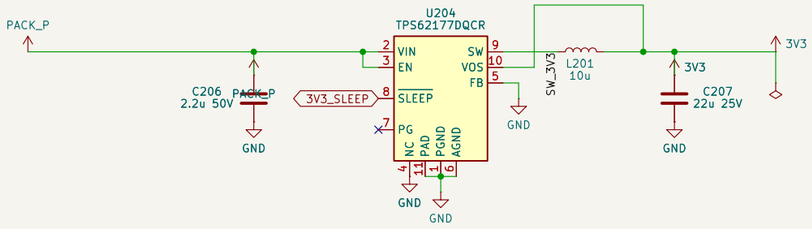



NOTES
1. CELL_N = gauge VSS (cell 1 negative). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads: R107/R108/C106 at the IC. RS101 is a 1206 1 W 75 ppm alloy part (P0).
3. VC taps are separate thin traces from the tap pads J109-J111 and from the J108/J112 pads:
   VC4 from CELL_P_RAW (before F102) so fuse drop is not read as cell voltage.
4. TS1-TS4: one wired 10k NTC per cell on J102, J105-J107, taped to the cell at mid-height; insulated sensors only
   (cans 0-12.6 V above VSS; TS 2.1 V max). PTC, PTCEN, PRES, DISP tied to VSS per datasheet.
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_P_RAW (cell top, before F102) via D101 + R105 (TI fig. 9-12).
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT onto the shared open-drain IRQ_N (CC2340 DIO11): IRQ_N is low while BTP_INT is high.
   Keep BTP disabled, or set BTP_POL = 1 with BTP_EN = 1, or IRQ_N is held low (review 4 D-2).
9. D103 clamps Q102 VGS at 15 V in a hard short. The dev board's C109/D102 inrush options are not on P0.
10. No wake button: the gauge leaves SHUTDOWN when PACK_P comes up, i.e. when USB is plugged in (TPS25751 dead-battery
    sink + BQ25756 charging with 3V3 down, USB sheet note 11). Confirm on the dev board.
11. TP110-TP112 = SMBus pogo pads (SMBC, SMBD, GND) for an EV2400 / BQStudio: golden image and calibration.
    Only with 3V3 up and U301 held in reset (MCU sheet note 10).
12. TP106 PACK_P and TP109 GND are factory test pads.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2). C107+C108 across PACK+/PACK- (sec. 9.2.2.1.5).
14. Second level (TI fig. 9-11): U102 BQ294700 (4.35 V/cell, CD 0.1u = 1-2 s) or the gauge FUSE pin turns Q104 on;
    F102's heater then blows its fuse, permanently opening the cell top. F102 is also the pack's only overcurrent fuse.
15. Connect the pack with USB unplugged, PACK- first, then the taps bottom-up, then PACK+: a floating tap can read
    > 4.35 V, U102 trips after 1-2 s and F102 blows. Gauge data flash: COV Delay 1 s, Min Blow Fuse Voltage 10500 mV.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
Welded 4S pack pads, BQ40Z50-R2 gauge/protector, protection FETs
**Monolith 100 W battery bank — production**

| | | |
|---|---|---|
| Sheet: /Cells and Gauge/ | | |
| File: gauge.kicad_sch | | |
| ***Title: Monolith production board — cells and gauge*** | | |
| Size: A3 | Date: 2026-10-17 | **Rev: P0** |
| KiCad E.D.A. 10.0.7 | | Id: 2/6 |

# 3.3 V ALWAYS-ON RAIL  (TPS62177, 1 MHz, sleep mode under MCU control)




NOTES
1. PACK_P -> TPS62177 -> 3V3 (MCU + both TPS25751). Gauge BAT is NOT on this rail (TI: gauge must stay powered with the FETs off).
2. SLEEP low = sleep mode (internal 1.4 uA pull-down); CC2340 DIO21 (3V3_SLEEP) drives it, and R313 220k holds it high while U301 is in reset or unprogrammed.
3. PG is not used.
4. U204 pin 4 (NC) goes to GND, as TI recommends.
5. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
6. The dev board's sleep-current section (R201 R_MEAS, 2 x INA190, HT7550-1) is not on P0: PACK_LP is PACK_P. Measure sleep current on the dev board.

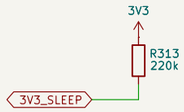
P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS62177 3V3 always-on rail
**Monolith 100 W battery bank — production**

Sheet: /3V3 Rail/
File: power.kicad_sch

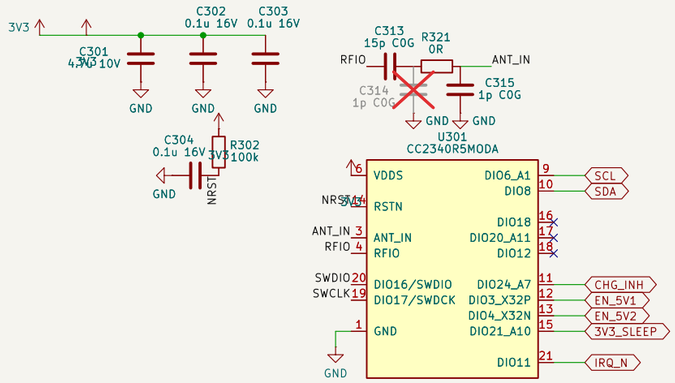
***Title: Monolith production board — 3V3 rail***

| Size: A3 | Date: 2026-10-17 | **Rev: P0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/6 |

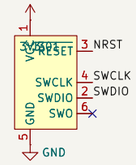
## 3V3_SLEEP PULL-UP



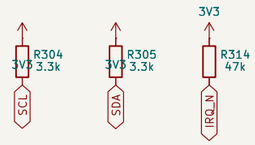

## BLUETOOTH MCU  CC2340R5MODA  (per TI SWRS349A fig. 9-1)




## SWD  (Tag-Connect TC2030-NL)




## I2C0 + IRQ PULL-UPS




NOTES
1. U301 antenna end at a board edge, ground 10-20 mm each side; keep-out in the footprint (fig. 9-2).
2. C301-C303 at the VDDS pads. R302 + C304 = power-on reset (SWRS349A fig. 9-1 R1/C92); no reset button.
3. C313 = CA1. R321/C314/C315 = Z61/Z60/Z62 pi match (TI: 0R, DNP, 1 pF); retune on the board.
4. No 32 kHz crystal: LFOSC calibrated against the 48 MHz crystal. DIO3/DIO4 are GPIOs.
5. Pins: DIO6 SCL, DIO8 SDA, DIO24 CHG_INH, DIO3/4 EN_5V1/2, DIO21 3V3_SLEEP, DIO11 IRQ_N.
   Spare (no-connect): DIO12, DIO18, DIO20, the dev board's LED and UART pins (docs/devboard/cc2340-pin-budget.md).
6. IRQ_N = TPS25751 IRQ x2 + gauge BTP (Q103), all open-drain. One pull-up: R314 47k.
   Firmware polls each source to find which.
7. Diamonds are PWR_FLAGs (ERC only, no part) where a rail is sourced through a passive part.
8. J301 = Tag-Connect TC2030-NL pads (no part fitted): XDS110 through a TC2030-CTX cable (10-pin Cortex to TC2030).
   SWO (pin 6) not used. The debugger drives NRST. Bootloader: ROM bootloader on a blank part or the CCFG backdoor pin.
9. R313 220k holds 3V3_SLEEP high in reset (pins float; TPS62177 pulls down 1.4 uA typ).
   Costs 15 uA while driven low in S2. DIO21 is also the default ROM-bootloader pin:
   disable or move that trigger in the CCFG before the first flash (review 4 A-1).
10. R304/R305 SMBus pull-ups go to 3V3. Gauge pogo-pad work (TP110-TP112) only with 3V3 up and U301 held in reset
    (NRST low through J301): an unpowered CC2340 can't take bus voltage (VDDS + 0.3 V max).
11. DIO24 = CHG_INH: LOW allows charging. High or hi-Z stops both chargers
    (R417/R517 pull CE up), so reset/hang/unprogrammed stops charging.
12. Watchdog: CC2340 WDT runs in active, idle and standby and can't be stopped once enabled.
    GPIOs latch in standby; IRQ_N wakes it on an edge (SWRS349A 8.6, 8.11).
13. Temps: gauge DAStatus2 (cells, FETs); charger NTCs via TPS25751 'I2Cr' (BQ25756 REG0x37).
14. TP301 3V3 is a factory test pad.

## FACTORY TEST PAD




P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
CC2340R5MODA, TC2030-NL SWD pads
**Monolith 100 W battery bank — production**

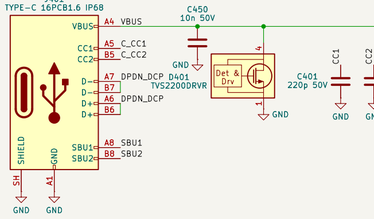
Sheet: /MCU/
File: mcu.kicad_sch

***Title: Monolith production board — MCU***

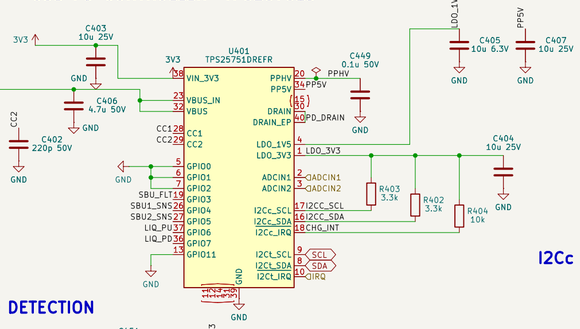
| Size: A3 | Date: 2026-10-17 | **Rev: P0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/6 |

Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

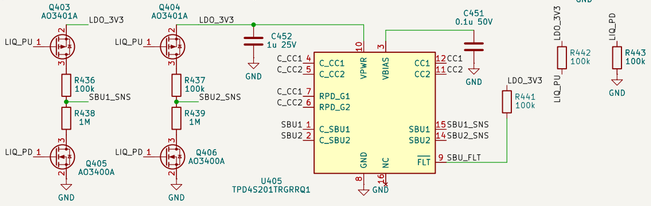
## USB-C CONNECTOR + PROTECTION



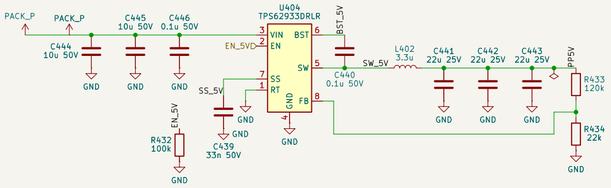

## USB PD CONTROLLER TPS25751DREFR




## CC/SBU PROTECTION (TPD4S201) + LIQUID DETECTION



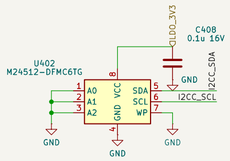

## PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN



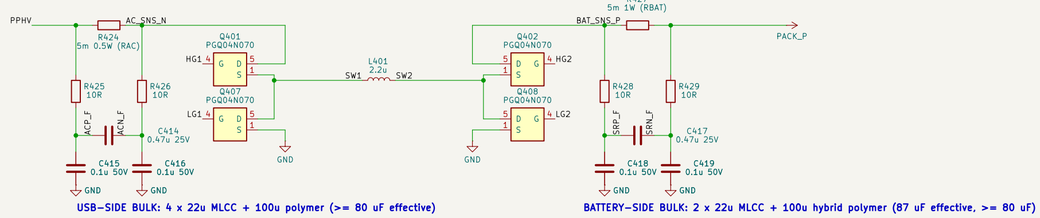

## FACTORY TEST PAD




## I2Cc BUS: EEPROM + PULL-UPS

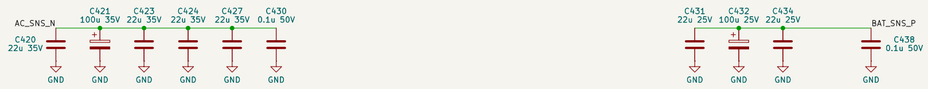



## BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

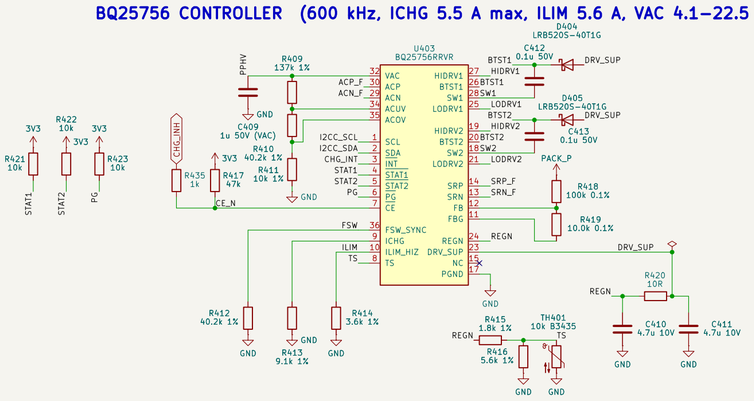



USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

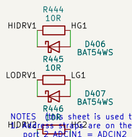
BATTERY-SIDE BULK: 2 x 22u MLCC + 100u hybrid polymer (87 uF effective, >= 80 uF)




## BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1–22.5 V window)




## GATE RESISTORS + TURN-OFF DIODES




NOTES — this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Strap pins are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751D GPIOs (0, 1, 2, 11) tied to GND per datasheet (KiCad ERC flags this: expected).
Never configure GPIOs as outputs.
3. CC and SBU are protected by U405 (TPD4S201, below): C401/C402 (220 pF) on the controller side.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the bulk caps (C421 polymer, C432 hybrid polymer) and remaining MLCCs
one row back. Polymer carries the bulk (no DC-bias loss), MLCCs carry the switching-frequency ripple.
Battery side is 2 x 22u + hybrid (C435 removed: docs/devboard/cap-reduction-sims-2026-10-17.md, B2: 87 uF,
hybrid at 63 % of its ripple rating). Still on hold: C427 (USB side, polymer at 85 % in the worst reverse case).
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly; take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD cSnkBulkPd: the TPS25751D sink switch soft-start limits
inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0).
Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH: R417 pulls CE up to 3V3, so charging needs the MCU (DIO24) to drive CHG_INH low through R435.
Reset, WDT reset, the ROM bootloader or an unprogrammed MCU stop charging; CC2340 Standby keeps the pin state.
A dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Gate drive: R444–R447 10R at the FET gates with D406–D409 for fast turn-off. BTST is wired straight to its cap
and there are no snubbers (removed in P0 before any bring-up data: add them back if the dev board shows ringing).
13. TP401 VBUS is a factory test pad.
14. PP5V output: C441–C443 3 x 22u 1206 (about 33 uF effective) plus C407 10u at the PP5V pin. Each port's PP5V is
its own TPS62933, a dedicated supply: USB PD cSrcBulk (10 uF min) applies, not the 120 uF cSrcBulkShared the
TPS25751 table cites. Estimated 0.5 -> 3 A step undershoot about 190 mV (cap sims P3); measure it on the dev board.
15. L401 = Coilank APS0730M2R2F (7.8 x 7.6 x 3.1 mm; dev board: 10 x 10 mm). 13.7 mOhm, Isat 17 A at a 40 % drop.
Worst case (20 V 5 A reverse from a 10 V pack): ~10.7 A rms, ~1.6 W DCR loss, L near 1.7 uH at the 12 A peak
(L/RSNS 0.34 uH/mOhm, inside TI's 0.05–1.26). Check its temperature on the dev board: the thermal throttle covers it.
16. RAC R424 is 0805 0.5 W (<= 0.16 W at 5.6 A). RBAT R427 1206 1 W (~0.5 W at 10 A). Both stay 5 mOhm, so ICHG/ILIM
settings are unchanged. Gate resistors R444–R447 and R420 are 0402; 0201 is used only where both nets stay <= 5.5 V.

PO: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
**Monolith 100 W battery bank — production**

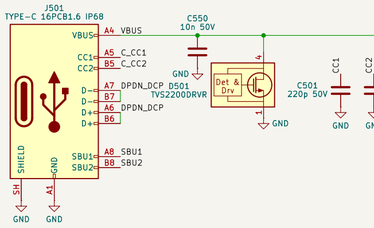
Sheet: /USB Port 1/
File: usb_port.kicad_sch

**Title: Monolith production board — USB-C port**

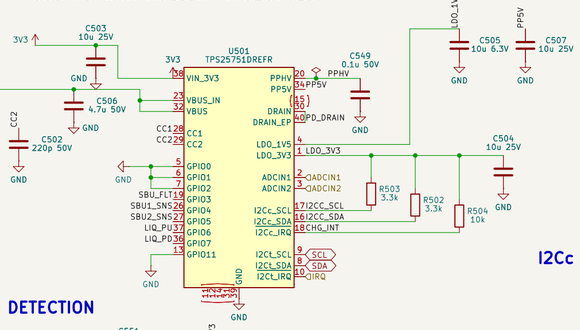
| Size: A2 | Date: 2026-10-17 | **Rev: P0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/6 |

Charger battery side joins PACK_P here (net tie R430 removed).
Route SRN (R429) and the FB divider top (R418) from R427's PACK_P pad,
not from the pour: DRC can't catch a tap on the bus.

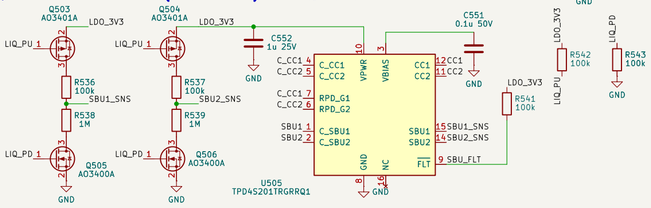
## USB-C CONNECTOR + PROTECTION



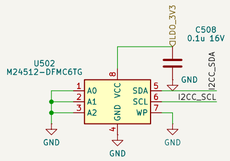

## USB PD CONTROLLER TPS25751D



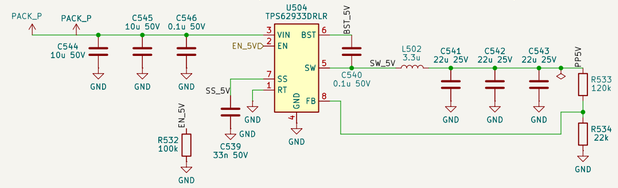

## CC/SBU PROTECTION (TPD4S201) + LIQUID DETECTION




## I2Cc BUS: EEPROM + PULL-UPS



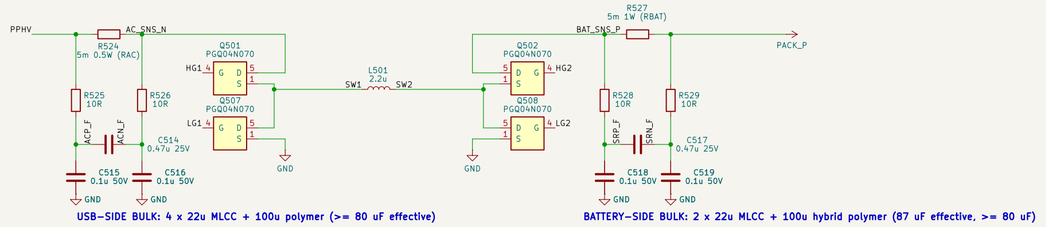

## PP5V SUPPLY TPS62933 (1.2 MHz, 5.16 V, 3 x 22u out) — 5 V source + VCONN




## FACTORY TEST PAD

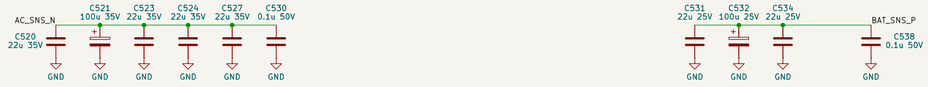



## BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

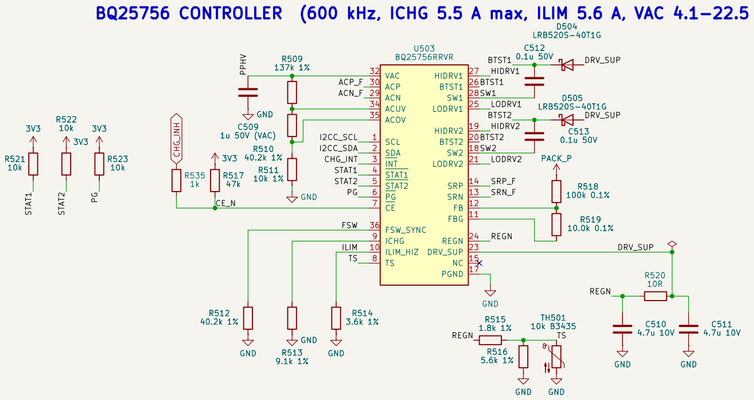



USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

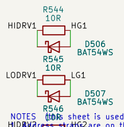
BATTERY-SIDE BULK: 2 x 22u MLCC + 100u hybrid polymer (87 uF effective, >= 80 uF)




## BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1–22.5 V window)




## GATE RESISTORS + TURN-OFF DIODES




NOTES
1. This sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
   Values are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
   port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751D PIOs (0, 1, 2, 11) tied to GND per datasheet (KiCad ERC flags this: expected).
   Never configure these as outputs.
3. CC and SBU are protected by U405 (TPD4S201, below): C401/C402 (220 pF) on the controller side.
   VBUS and GND are tied together at the connector (net DPDN_DCP: passive BC1.2 DCP): the TPS25751 does no BC1.2.
   VBUS detection (TI fig. 9-8): GPIO4/5 read SBU1/2 through U405 (TPD4S201: CC and SBU OVP + ESD);
   GPIO6 = liquid_pmos_control (LIQ_PU), GPIO7 = liquid_nmos_control (LIQ_PD) in the TPS25751 configuration.
   U405 pin 5 = !OVP_FLT = Fault_Input_Event (SBU_FLT, R441); an SBU short to VBUS forces error recovery.
   R442/R443 hold the liquid-detect FETs off while GPIO6/7 are Hi-Z.
   D401 = TVS2200 flat clamp on VBUS (28 V at 40 A 8/20 us; TI's pick for the TPS25751).
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the bulk caps (C421 polymer, C432 hybrid polymer) and remaining MLCCs
   one row back. Polymer carries the bulk (no DC-bias loss), MLCCs carry the switching-frequency ripple.
   Battery side is 2 x 22u + hybrid (C435 removed: docs/devboard/cap-reduction-sims-2026-10-17.md, B2: 87 uF,
   hybrid at 63 % of its ripple rating). Still on hold: C427 (USB side, polymer at 85 % in the worst reverse case).
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. The charger battery side is PACK_P directly; take SRN/FB from R427's pad, not from the pour.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD cSnkBulkPd: the TPS25751D sink switch soft-start limits
   inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0).
   Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH: R417 pulls CE up to 3V3, so charging needs the MCU (DIO24) to drive CHG_INH low through R435.
    Reset, WDT reset, the ROM bootloader or an unprogrammed MCU stop charging: CC2340 Standby keeps the pin state.
    A dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Gate drive: R444-R447 10R at the FET gates with D406-D409 for fast turn-off. BTST is wired straight to its cap
    and there are no snubbers (removed in P0 before any bring-up data: add them back if the dev board shows ringing).
13. TP401 VBUS is a factory test pad.
14. PP5V output: C441-C443 3 x 22u 1206 (about 33 uF effective) plus C407 10u at the PP5V pin. Each port's PP5V is
    its own TPS62933, a dedicated supply: USB PD cSrcBulk (10 uF min) applies, not the 120 uF cSrcBulkShared the
    TPS25751 table cites. Estimated 0.5 -> 3 A step undershoot about 190 mV (cap sims P3); measure it on the dev board.
15. L401 = Coilank APS0730M2R2F (7.8 x 7.6 x 3.1 mm; dev board: 10 x 10 mm). 13.7 mOhm, Isat 17 A at a 40 % drop.
    Worst case (20 V 5 A reverse from a 10 V pack): ~10.7 A rms, ~1.6 W DCR loss, L near 1.7 uH at the 12 A peak
    (L/RSNS 0.34 uH/mOhm, inside TI's 0.05-1.26). Check its temperature on the dev board: the thermal throttle covers it.
16. RAC R424 is 0805 0.5 W (<= 0.16 W at 5.6 A). RBAT R427 1206 1 W (~0.5 W at 10 A). Both stay 5 mOhm, so ICHG/ILIM
    settings are unchanged. Gate resistors R444-R447 and R420 are 0402; 0201 is used only where both nets stay <= 5.5 V.

P0: dev board rev A8 stripped by Production/tools/make_production.py; not reviewed
Spec + open items: Production/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
**Monolith 100 W battery bank — production**

Sheet: /USB Port 2/
File: usb_port.kicad_sch

**Title: Monolith production board — USB-C port**

| Size: A2 | Date: 2026-10-17 | **Rev: P0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/6 |